# UofT Phishnet

## A Solution for UofT Phishing Email Problems

MIE1517 Introduction to Deep Learning: Group 11 Final Presentation

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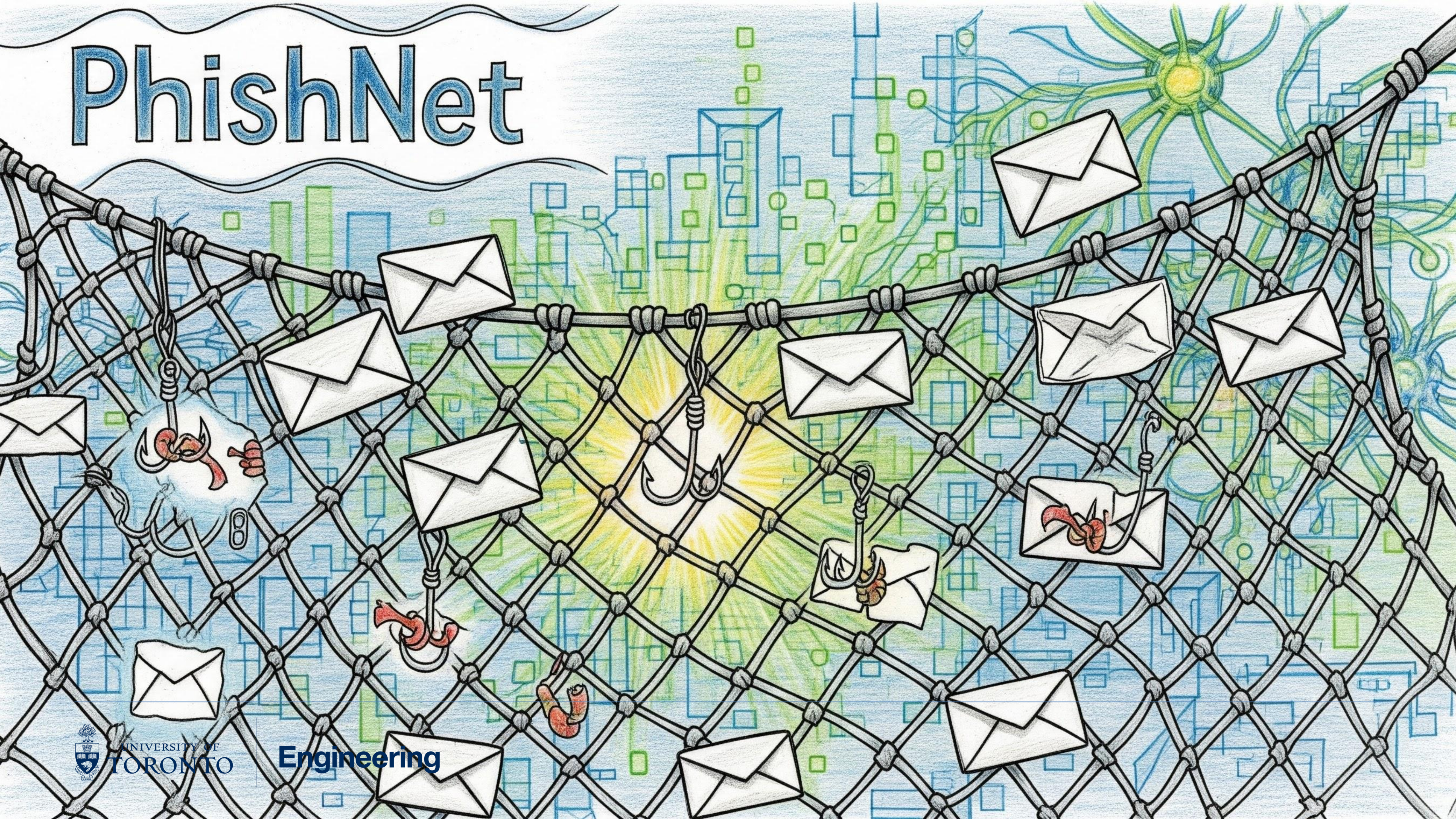
[a]Department of Mechanical and Industrial Engineering (MIE)
[b]University of Toronto Institute for Aerospace Studies (UTIAS)
**University of Toronto**


PhishNet
UNIVERSITY OF TORONTO
Engineering

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# Introduction

## Motivational Background

EK Eunjin Ko

Reply | Reply all | Forward

Mon 08/09/2025 17:58

You replied on Mon 08/09/2025 19:32

Dear Student,

We are writing to remind you that our records indicate your **Fall 2025 tuition fees remain unpaid**. As per the **University of Toronto's Financial Aid & Registration Policy**, an **initial deposit of $2,000–$5,000 is required by today, September 8, 2025**, in order to secure your course registration and maintain uninterrupted access to academic services.

To complete your payment, please send an **Interac e-Transfer** to the University's BACPURSE account using the following details:

**Email: Nckamogu97@gmail.com**

**Message/Memo:** Please include your student number

This initial deposit is **mandatory**, even if you have applied for **OSAP or are awaiting external funding**. OSAP disbursements are typically processed **after the term begins**, and therefore do not exempt students from the upfront deposit requirement.

Failure to submit the initial deposit today may result in:

- Removal from registered courses
- Holds on academic records (transcripts, graduation eligibility)

# Introduction

## Motivational Background

[redacted] • 3mo ago

I WAS ABOUT TO SEND THE MONEY but got sus and came to ask here!!! THANK GOD and thank youuu

0

[redacted] • 3mo ago

I got this exact email on Aug 30th saying they're gonna withdraw me from my courses because I still owed tuition. I did in fact still owe tuition and automatically assumed this was true because of the timing (fall deadline fees) and utoronto.ca email. I ended up sending 3k. Looking back it was so obvious but in the moment I didn't even think to check if it was fraud. I've never felt so stupid in my life and I'm genuinely not sure if I'll be able to leave my bed. Obviously I've contacted my bank, the police, and campus security but I'll prob not get the money back and will have to pay my tuition (3k) to utm eventually. Idc if people make fun of me, I made a mistake and I'll just have to deal with the consequences.

2

# Introduction

## Motivational Background

[redacted]
Associate Director, Information Security Strategic Initiatives

SEP 19

**Marc Bishara** • 9:35 AM

Hi [redacted],

Thank you for accepting my request. I got your contact from [redacted]. I am an MEng student enrolled in MIE. I am working along with three other colleagues (all PhD candidates) on a project for efficient classification of phishing emails as part of a deep learning class: MIE1517 with Prof Sinisa Colic. We were wondering if there could be room for collaboration with your department in our project. If that is something of interest to you, please let me know and I would be happy to share more details.

Best Regards
Marc Bishara
marc.bishara@mail.utoronto.ca

[redacted] • 9:50 AM

Hi Marc. We would be happy to help. What kind of collaboration are you looking for?

---

[redacted]

Reply | Reply all | Forward

To: Marc Bishara

Cc: [redacted]

Tue 18/11/2025 16:49

You replied on Tue 18/11/2025 17:24

Hi Marc,

Thank you for your email. Yes this request has been brought to my attention and I am looping back [redacted] in the email chain for visibility.

Please be advised we have reached out internally to the respective team to identify the level of data that is available for us to share and we shall reach back to you once we have more information on this.

Thanks

[redacted]

[redacted]

IS Risk Manager | Information Security| ITS

For General Risk related enquiries, please raise a ticket at Get Help

...

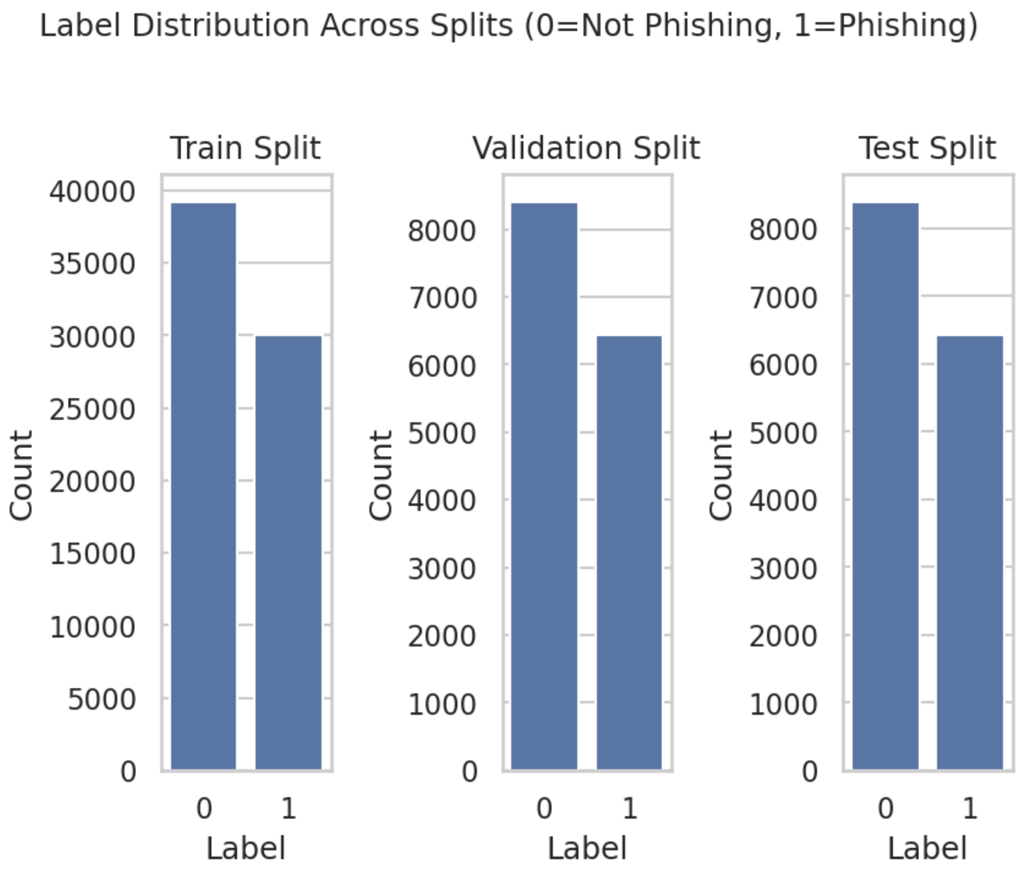
# Dataset

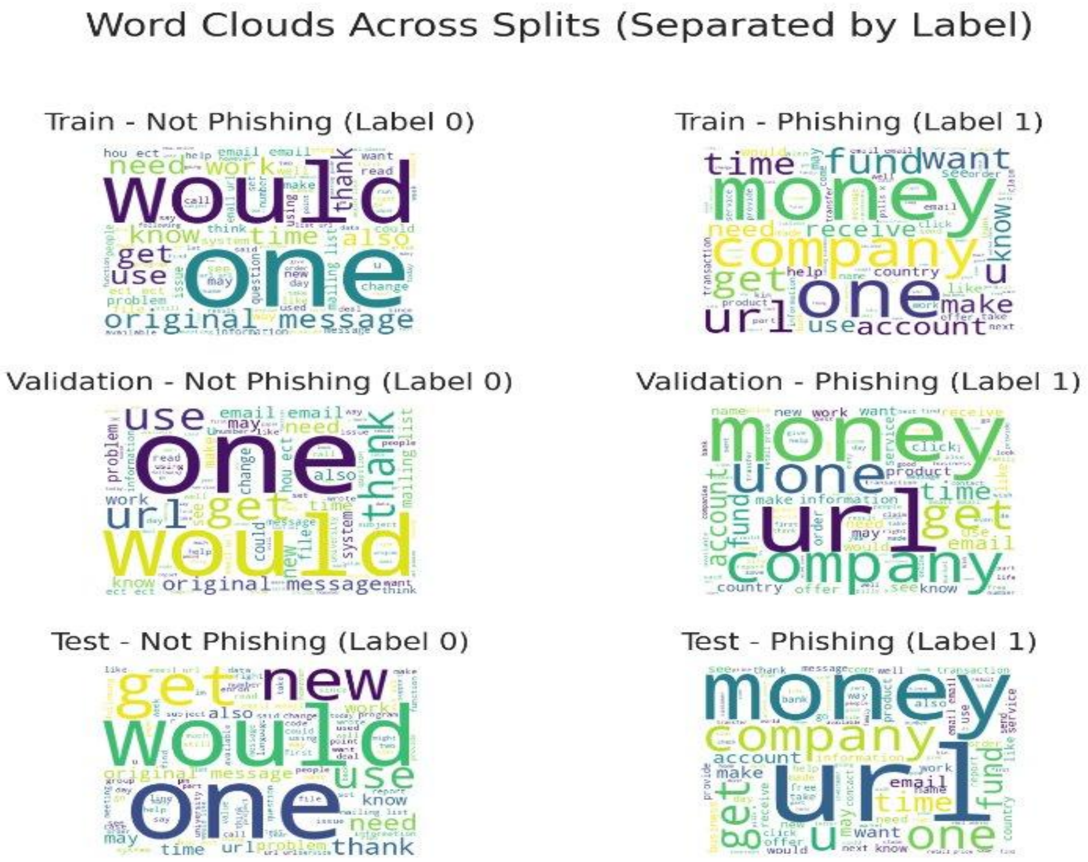
| Source | Email count |
|---|---|
| Arifa Islam, C. (2023) | 203K over 11 datasets |
| Chakraborty, S. (2023) | 18K |
| Miltchev, R. (2025) | 2K |
| Radev, D. (2008) | 11.5K |
| Hunter Biden dataset | A subset of 1K safe emails |
| Synthetic dataset | 242 |

Total 235,742 from which we sampled 70,000 for training; 15,000 for validation and 15,000 for test

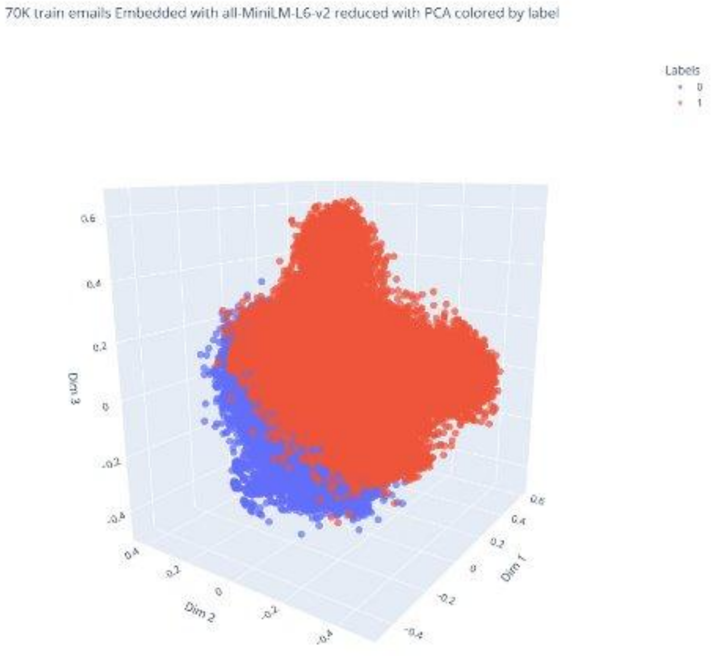
# Dataset - EDA

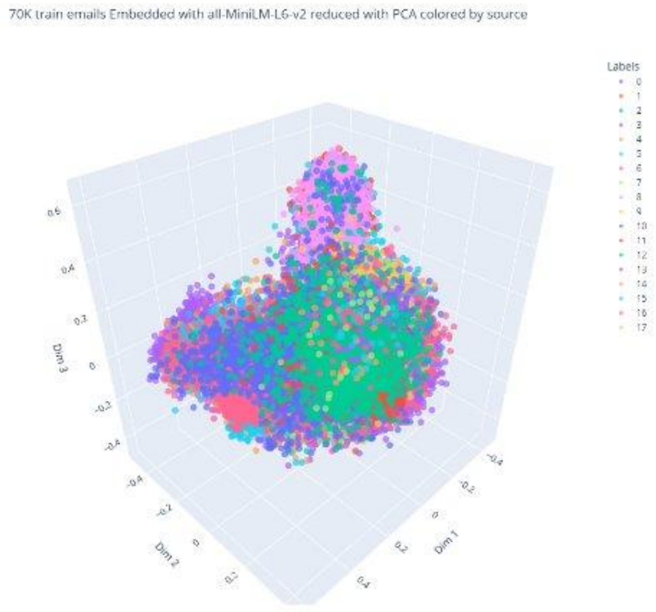
- Train: 70,000 | Validation: 15,000 | Test: 15,000

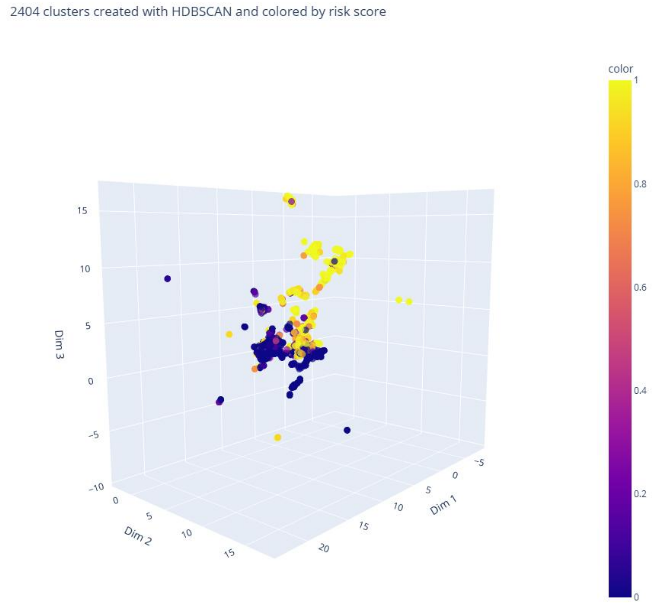


# Dataset – PCA Analysis



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# End-to-End Pipeline

# End-to-End Pipeline

Input:

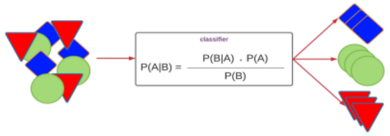
Phishing
+ Non-Phishing
Emails

ML Data
Preprocessing
Pipeline

DL Data
Preprocessing
Pipeline

## Machine Learning Models

### Naïve Bayes

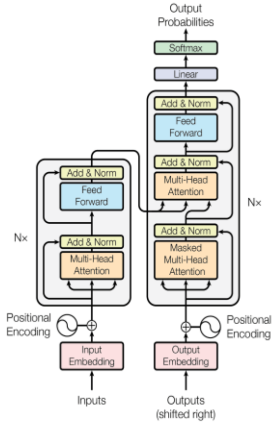



### Phish score

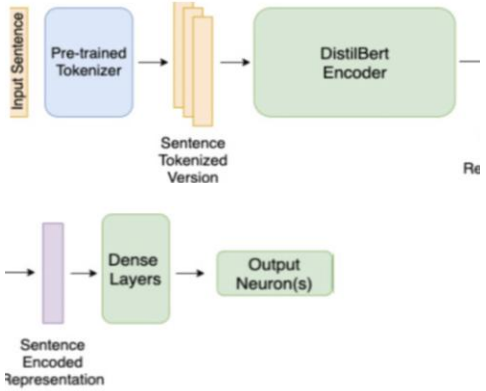
SBERT.net
HDBSCAN
UMAP

## Deep Learning Models

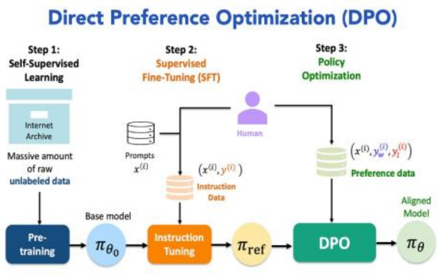
### BERT Variants




### DistillBERT




### Llama DPO



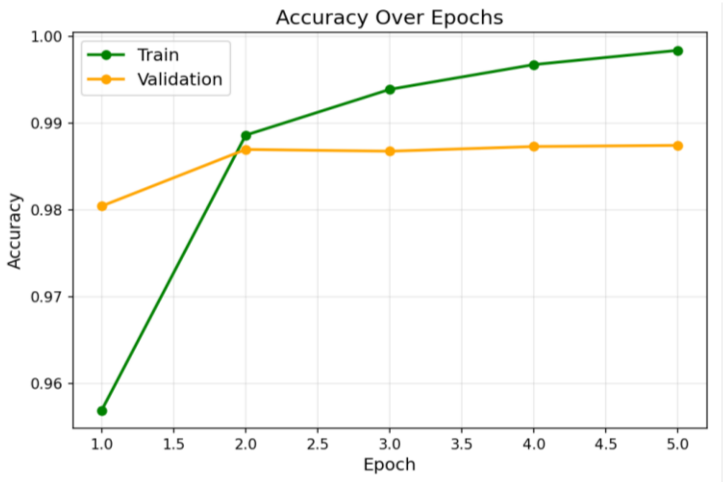

Output:

Bait or
Great!

# Results

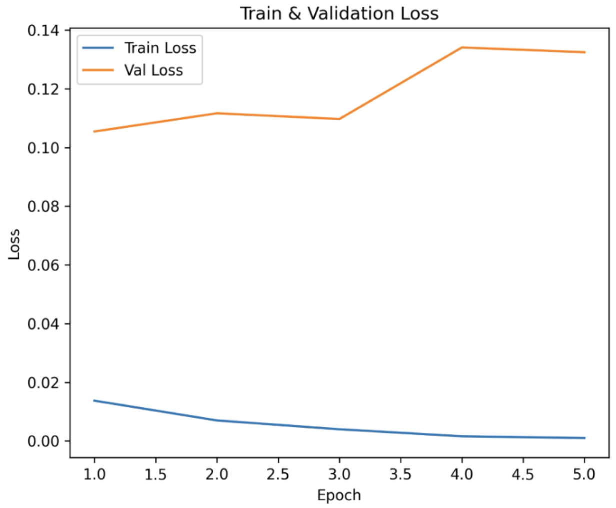
# Results - Training

## RoBERTa



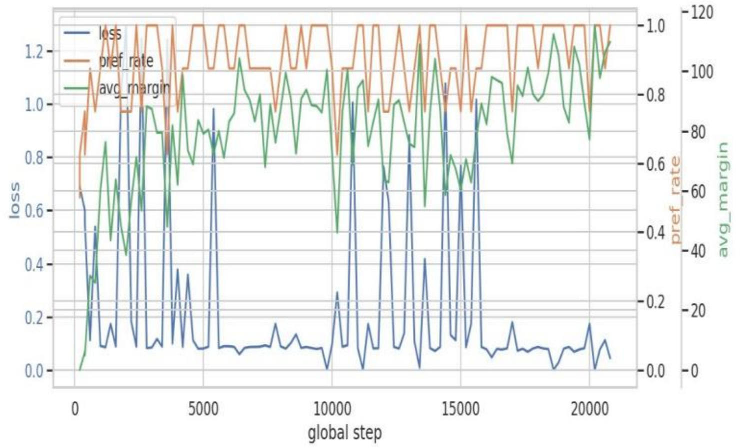

Training Epochs: 5
Best Validation Accuracy: 0.989

## DistilBERT




Training Epochs: 5
Best Validation Accuracy: 0.983

## Llama 3.2



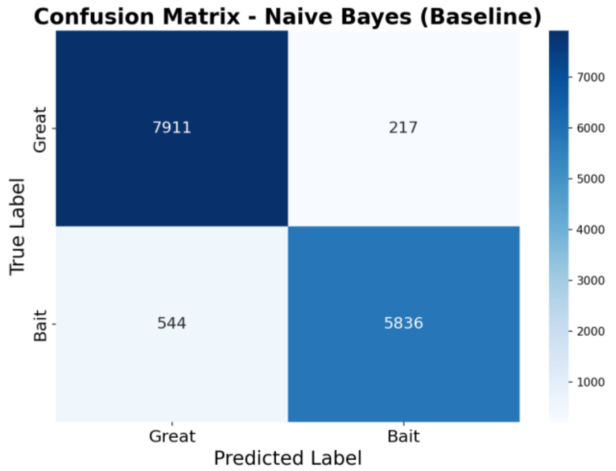

Training Epochs: 3
Best Validation Accuracy: 0.989

# Results – Test Inference Results

Naïve Bayes Model

Accuracy: 94.75%

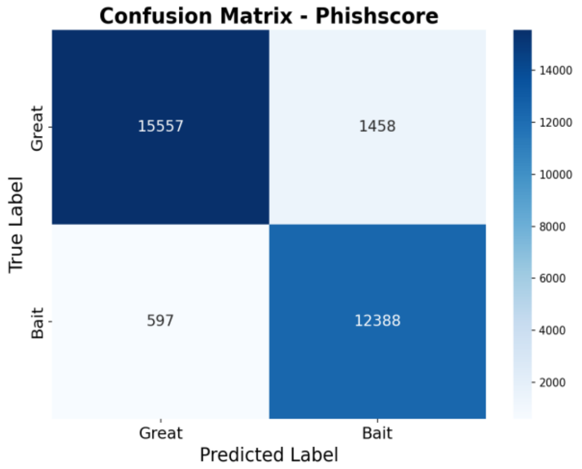
Weighted
Precision: 94.82%




Phish Score Model

Accuracy: 94.0%

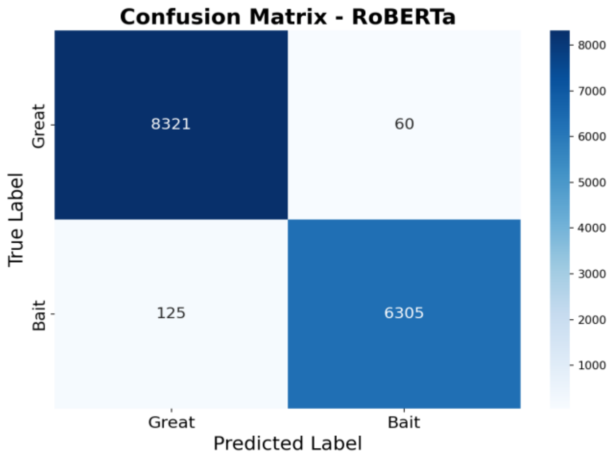
Weighted
Precision: 93.35%




RoBERTa Model

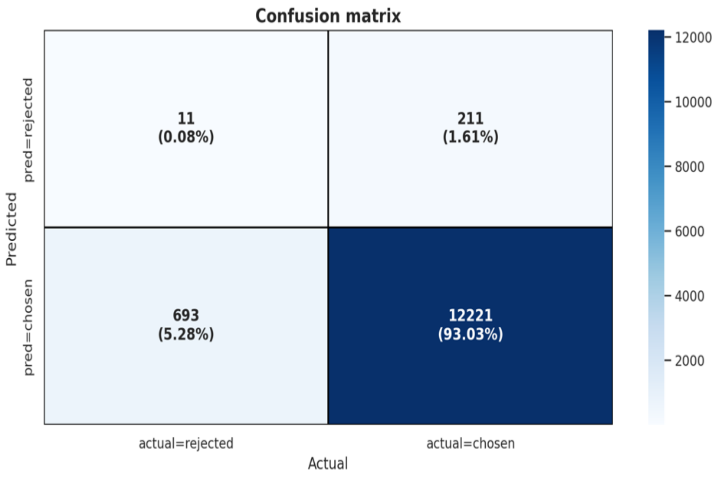
Accuracy: 98.74%

Weighted
Precision: 98.75%




Llama with DPO

Accuracy: 93.3%



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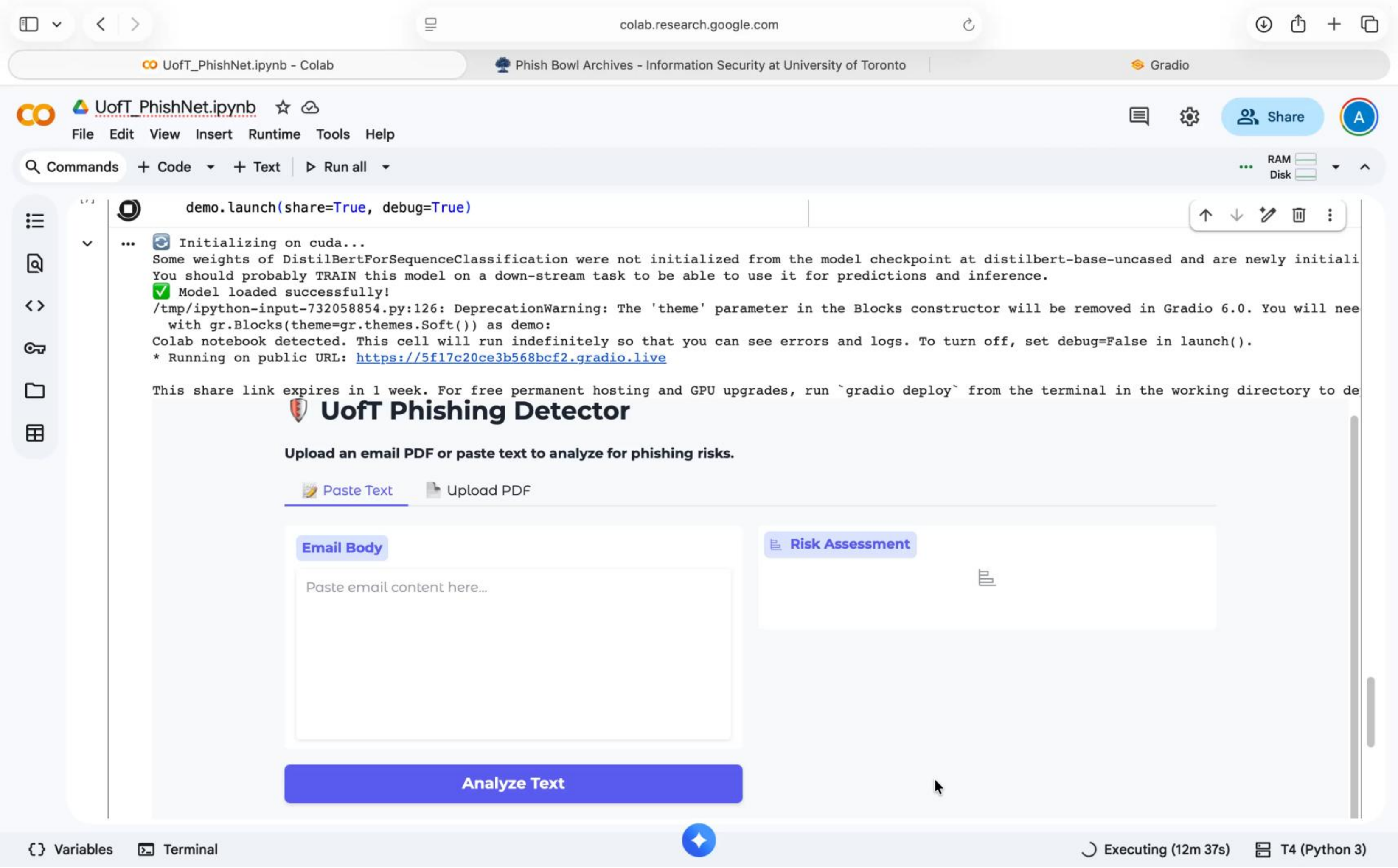
# Demonstration

# Demonstration



# Discussion – Summary of Approaches

| Model Name | Model Type | Phishnet Dataset Test Accuracy (%) | UofT Phishbowl Dataset Test Accuracy | Weighted Precision (%) | Num Parameters |
|---|---|---|---|---|---|
| **RoBERTa** | **DL** | **98.75** | 64.29 | **98.75** | 124.6M |
| Llama 3.2 1B - DPO | DL | 91.7- 98.5 | 35.9 | * | **1.2B** |
| BERT | DL | 98.47 | 64.29 | 98.47 | 108.3M |
| DistillBERT | DL | 98.44 | 71.43 | 98.44 | 65.8M |
| Phish score | Hybrid | 94 | 63.89 | 93.35 | 22.7M |
| Linear SVC (SVM) | ML | 97.2 | 71.43 | 97.2 | 20.0K |
| Logistic Regression | ML | 96.91 | 71.43 | 96.91 | 20.0K |
| SGD Classifier (SVM-like) | ML | 96.72 | 35.71 | 96.72 | 20.0K |
| Naive Bayes (Baseline) | ML | 94.75 | **85.71** | 94.82 | 40.0K |

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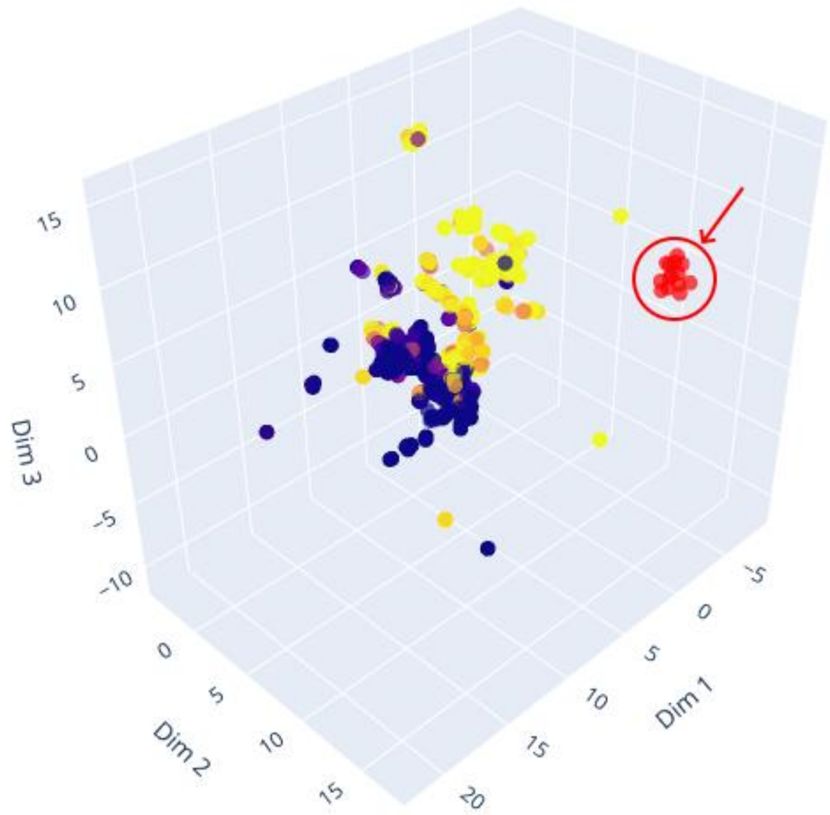
# Discussion

# Discussion – Demonstration data in latent space

- Demonstration emails are out of distribution

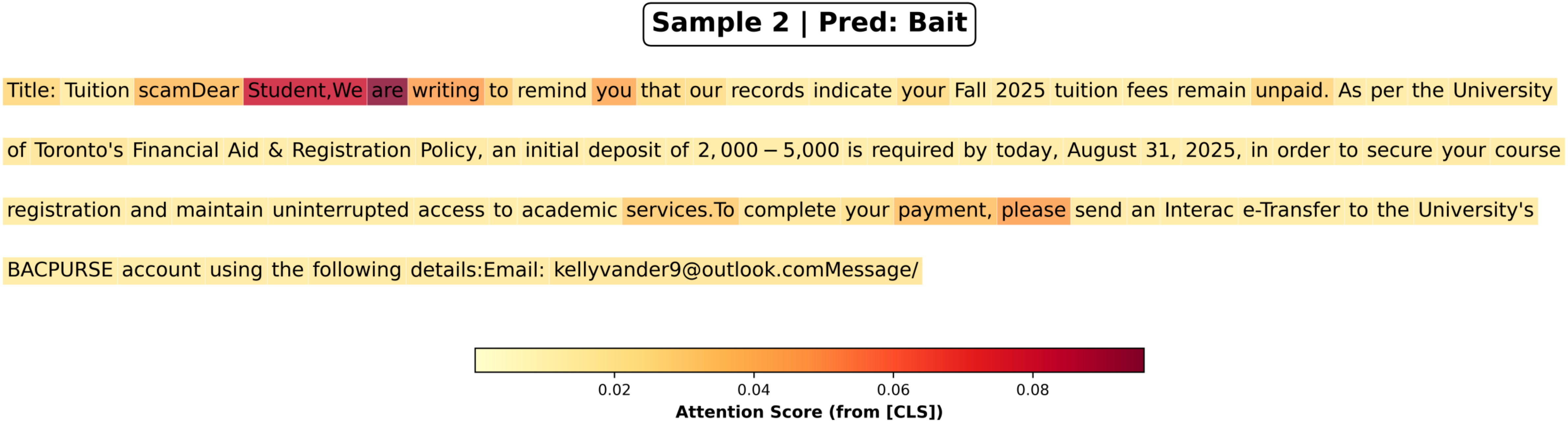



2404 clusters colored by risk score and 14 colored in red

Labels
Clusters, Blue = Great Yellow = Bait
Demo Emails Embeddings

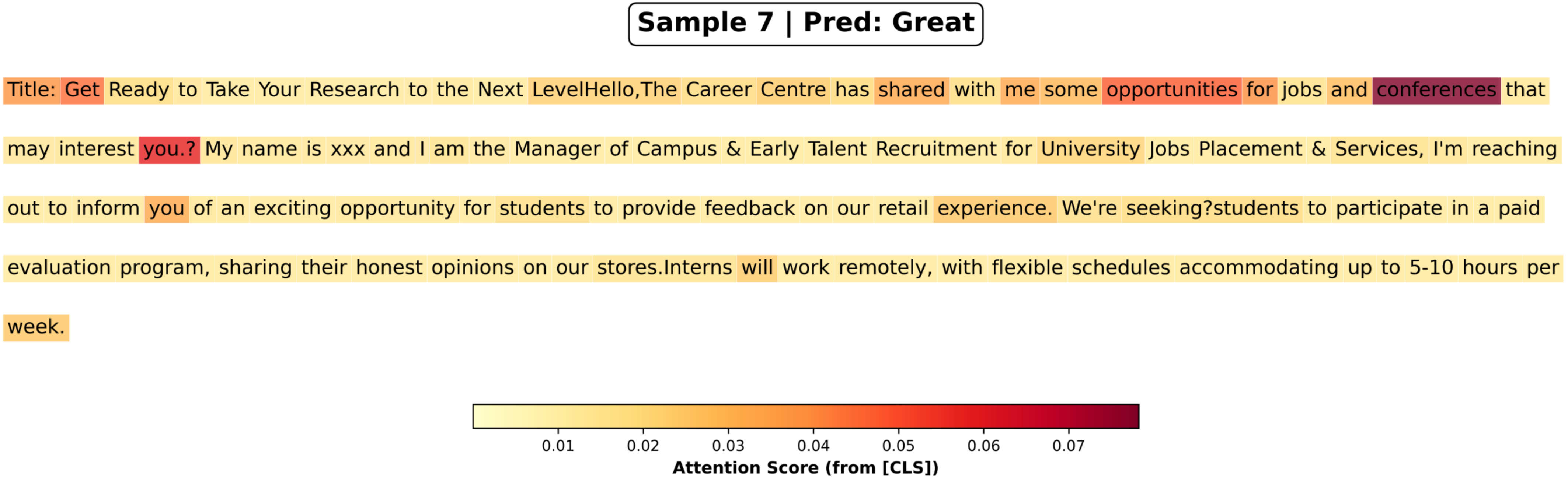
# Discussion – Model Interpretability

- Attention scores for emails predicted **correctly** – Phishbowl dataset



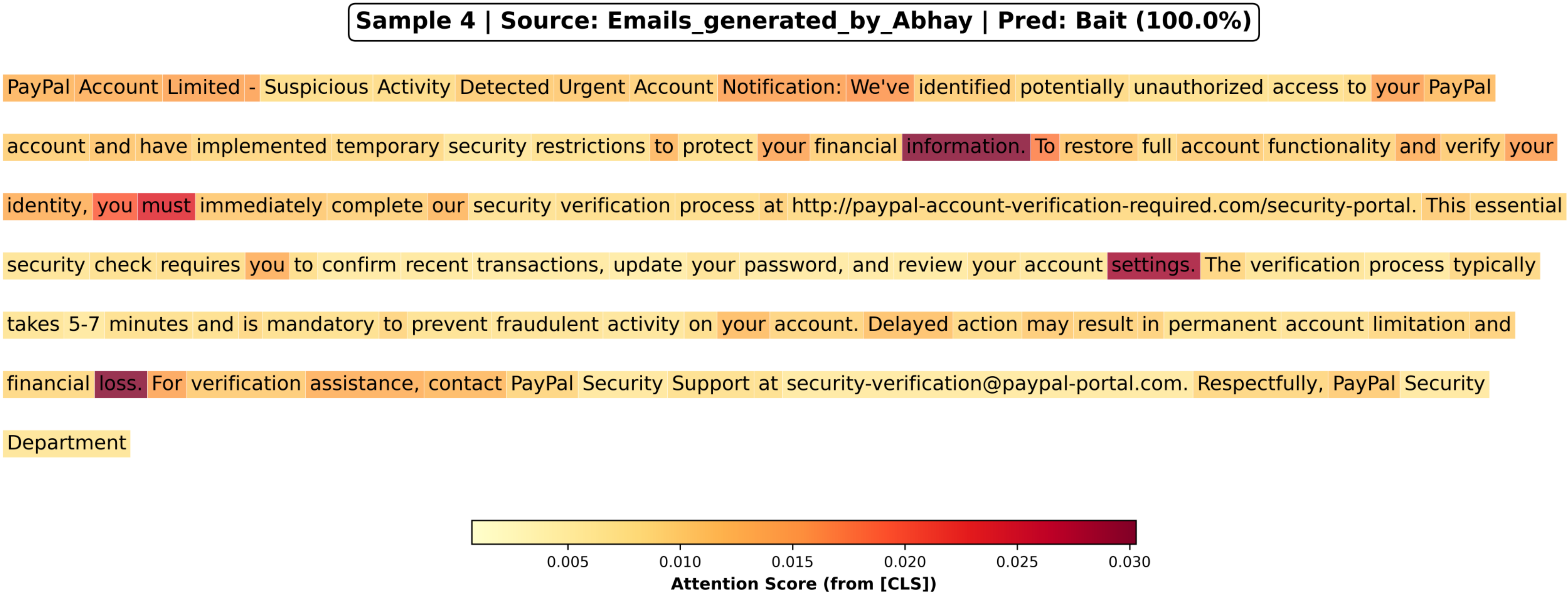
# Discussion – Model Interpretability

- Attention scores for emails predicted **incorrectly** – Phishbowl dataset



# Discussion – Model Interpretability

- Attention scores for emails predicted **correctly** – Synthetic dataset



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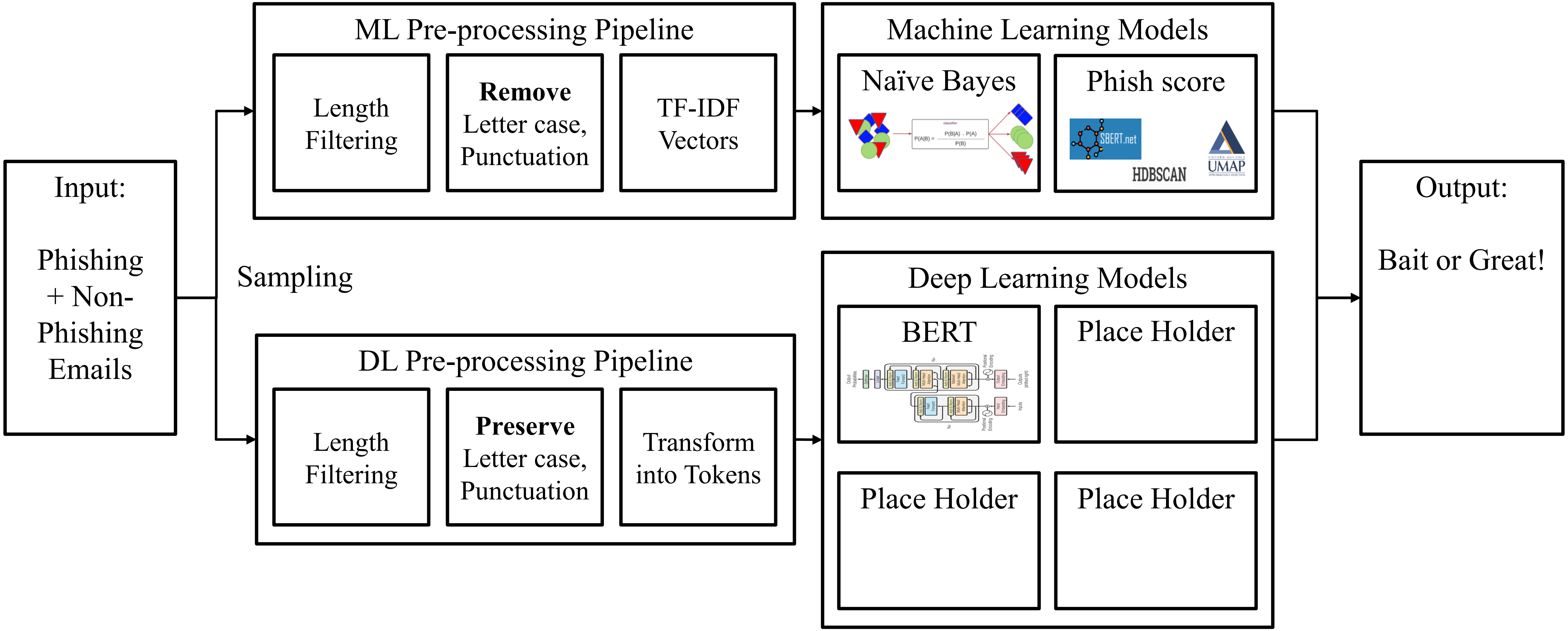
# Thank you.

# End-to-End Pipeline



# End-to-End Pipeline